

Worksheet · Section 5.1–5.2

Beneficial & Harmful Effects · Digital Divide



The Big Picture

Every computing innovation helps some people and can hurt others, and not everyone has equal access. Practice analyzing effects, naming digital-divide factors, and proposing solutions.

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|---|--|
| _____ 1. Computing innovation | A. Related to income and education |
| _____ 2. Beneficial effect | B. Unequal access to computing devices and the Internet |
| _____ 3. Harmful effect | C. An effect the creators did not plan for |
| _____ 4. Unintended consequence | D. Related to traits like age |
| _____ 5. Responsible programming | E. Considering possible misuses and harms of a program |
| _____ 6. Digital divide | F. An outcome that helps people or society |
| _____ 7. Socioeconomic | G. An outcome that hurts people or society |
| _____ 8. Geographic | H. The skills to use technology effectively |
| _____ 9. Demographic | I. Related to location, like rural vs. urban |
| _____ 10. Equity | J. An innovation that includes a program as an essential part |
| _____ 11. Digital literacy | K. Fairness — everyone gets what they need to succeed |

Part 2 · Fill It In

Use each word from the word bank once.

geographic**both****intended****socioeconomic****unintended****governments****literacy****digital divide**

1. An effect the creators didn't plan for is _____.
2. One innovation can have _____ beneficial and harmful effects.
3. Innovations can be used in ways their creators never _____.
4. Unequal access to computers and the Internet is the _____.
5. No broadband in a rural town is a _____ factor.
6. Not being able to afford a laptop is a _____ factor.
7. Individuals, organizations, and _____ can affect the digital divide.
8. Knowing how to use technology well is digital _____.

Name:

Class:

Date:

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Practice

Part 3 · Beneficial, Harmful, or Both?

Check the best description of each effect.

Item	Beneficial	Harmful	Both
1. GPS apps help drivers avoid traffic			
2. Online scams target older adults			
3. Telehealth lets rural patients see doctors			
4. Social media keeps families connected but spreads misinformation			
5. Self-checkout saves time but reduces cashier jobs			
6. Cyberbullying on messaging apps			
7. Free online courses teach new skills			
8. Facial recognition finds missing people but enables surveillance			

Part 4 · Name the Factor

Situation	Factor
A rural town has no broadband	_____
A family can't afford a data plan	_____
Grandparents are unfamiliar with apps	_____
A country blocks many websites	_____
A school lends Wi-Fi hotspots	_____

Part 5 · Innovation Analysis

Innovation	Intended purpose	Beneficial effect	Harmful or unintended effect
Smartphone cameras			
Online maps			
Your choice:			

Part 6 · Close the Gap

A district assigns homework that requires streaming video every night. Describe who is disadvantaged, then propose a plan with one action by the school and one by the government.

**Think:**

Use the “Who?” test: Who benefits? Who might be harmed? Who doesn't have access at all?

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. What is the digital divide?
- A. The gap between hardware and software
 - B. Unequal access to computing devices and the Internet
 - C. Splitting a program into parts
 - D. The difference between binary and decimal

★★ On Level

2. The Web was created so scientists could share research. Today it's used for shopping and entertainment. This shows that...
- A. the Web failed
 - B. the Web has only harmful effects
 - C. innovations can be used in ways their creators didn't intend
 - D. scientists no longer use the Web

★★ On Level

3. Which action would most directly reduce the digital divide?
- A. Libraries offering free computers and Wi-Fi
 - B. Raising Internet prices
 - C. Requiring all forms to be online only
 - D. Releasing a more expensive phone

★★★ Exam Hard

4. Ride-share apps reduced drunk driving but increased city traffic. Which best describes this?
- A. A syntax error
 - B. One innovation with both beneficial and harmful effects, some unintended
 - C. The digital divide
 - D. Only an intended benefit

★★★ Exam Hard

5. Which TWO statements are true? (*Select two.*)
- A. Everyone with a phone has equal access to technology.
 - B. Government policies can widen or narrow the digital divide.
 - C. The digital divide is only about owning hardware.
 - D. The digital divide exists within countries, not just between them.

Part 8 · Written Response

Pick an app you use every day. Describe one beneficial effect, one harmful effect, and a group that might be left out because of the digital divide.
